HSLU Hochschule Luzern

Data-driven Supply Chain Management & Logistics

Inventory Optimisation

26% less capital · 95% availability



Group 2 · Topic 1 · by Divya Shori

25 November 2026 · github.com/divyashori-datascience/scm-inventory-optimisation

EXECUTIVE SUMMARY

Free 1.27M of capital while raising availability

−26%
average stock vs 2021

95%
fill rate, 97% on A-articles

−34%
capital vs today’s rules, same service

WHAT WE RECOMMEND

1. Clean the range
2. Stock by class
3. Set levels from demand

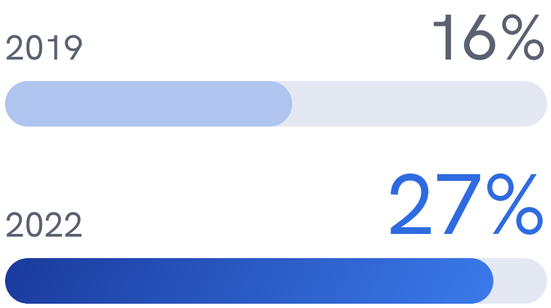
Simulated test year 2021 · 1,751 inventory articles

THE CHALLENGE & OUR APPROACH

Capital lockup rose from 16% to 27% of revenue

Three questions, each building on the one before

Capital in stock, % of revenue



+11 points in three years

1

PART 1 · DATA ANALYSIS

Where does the capital sit?

2

PART 2 · CLASSIFICATION

Which articles belong on stock?

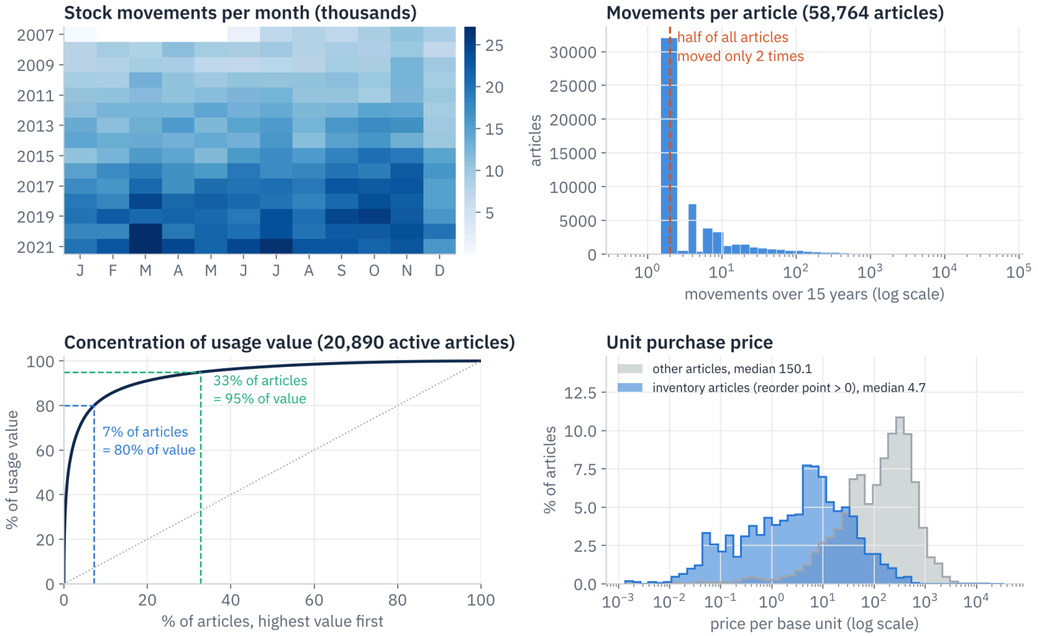
3

PART 3 · REORDER POINTS

At what level should they reorder?

Growing activity, a long tail, value in few articles

Why: understand the data before modelling it



more movements than in 2008

5×

of articles moved only twice

50%

of articles = 80% of value

7%

All 2,694,849 movements · value on the 20,890 articles with demand in the last 3 years

PART 1 · DATA PREPARATION

From 2.69M raw rows
to two clean analysis tables

Why: every later number rests on this pipeline

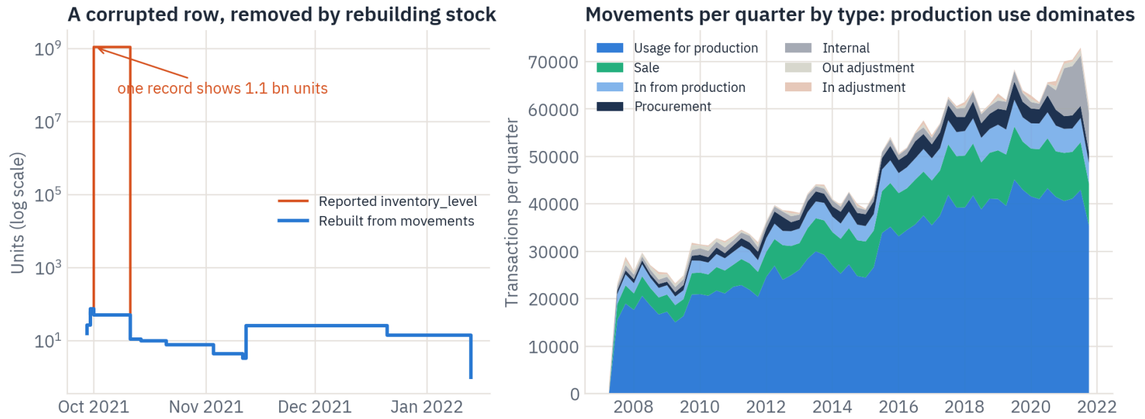


Code: shared/prep.py · quantities ×1.1107 (anonymised) · master data = 2022 snapshot

PART 1 · DATA CLEANING

One bad row would have put capital at 2.9 bn

Why: proof that the stock history had to be rebuilt



units in one corrupted row

1.1 bn

of movements are production use

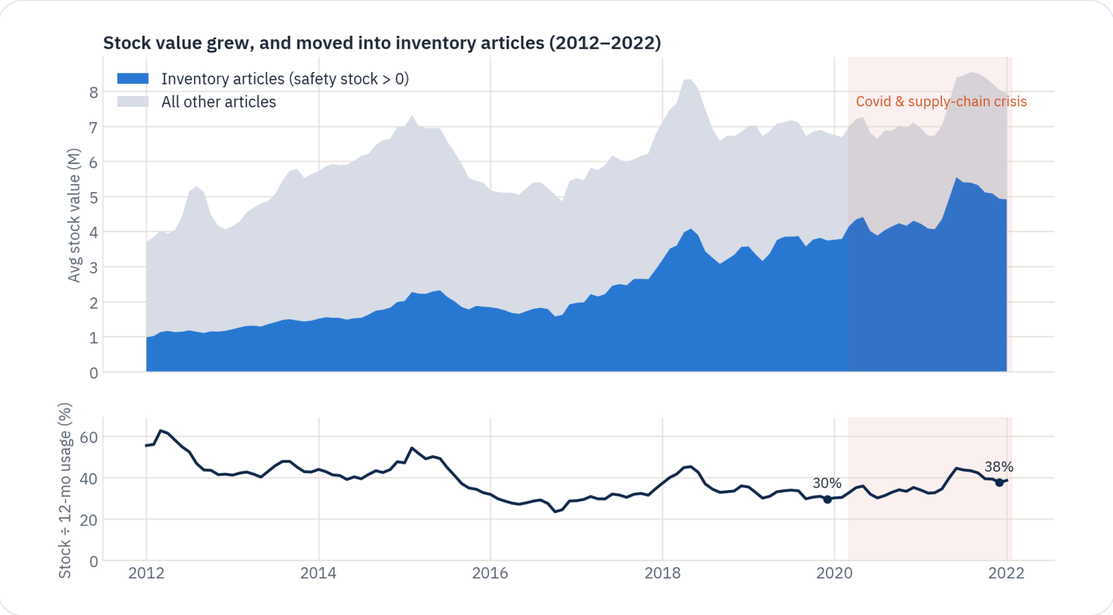
65%

Rebuilt stock = opening level + cumulative movements

PART 1 · WHERE THE CAPITAL SITS

Stock grew faster than demand, and a third is unmanaged

Why: shows where the money is tied up



stock in inventory articles since 2018

+39%

of a year's usage held as stock

30% → 38%

of stock has no reorder point

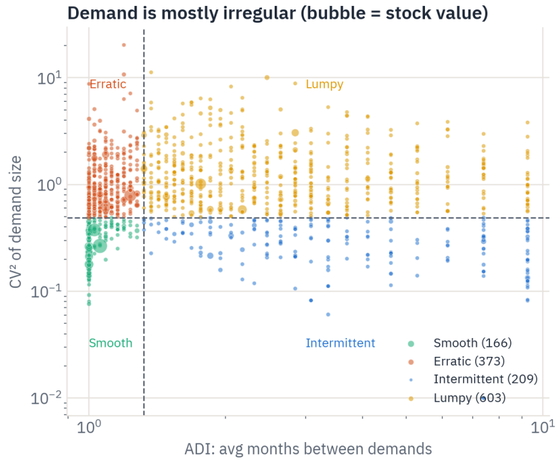
37%

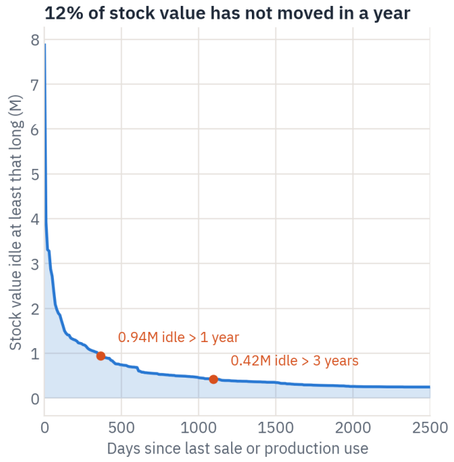
Inventory articles = 1,751 articles with a reorder point above zero

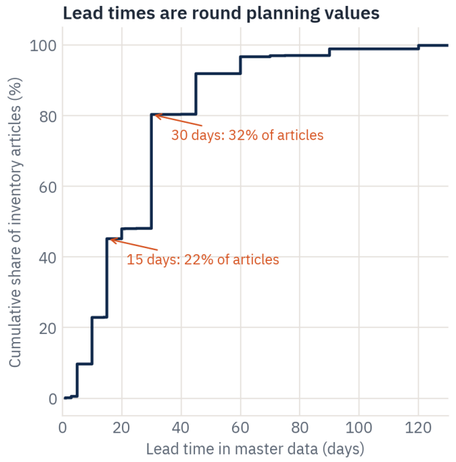
PART 1 · DEMAND, IDLE STOCK, LEAD TIMES

One formula cannot fit this range

Why: these three facts shape the solution







90%
of stock articles have irregular demand

0.94M
idle for more than a year

54%
of lead times are exactly 15 or 30 days

HOW WE CLASSIFY ARTICLES

Three classifications, three decisions

Why: one grouping cannot answer three different questions

1 · CLUSTERS · PART 2

Stock it at all?

Runners · Repeaters · One-offs · Dormant

→ **stock, to order or phase out**

2 · DEMAND PATTERN · PART 3

Which formula?

Smooth · Erratic · Intermittent · Lumpy

→ **how the reorder point is computed**

3 · ABC · PART 3

How much buffer?

A · B · C by usage value

→ **fill target 98% / 95% / 90%**

HOW ONE ARTICLE FLOWS THROUGH THEM

Article
any of 20,890

→

Cluster
stock it? if yes →

→

Pattern
pick the formula

→

ABC
set the target

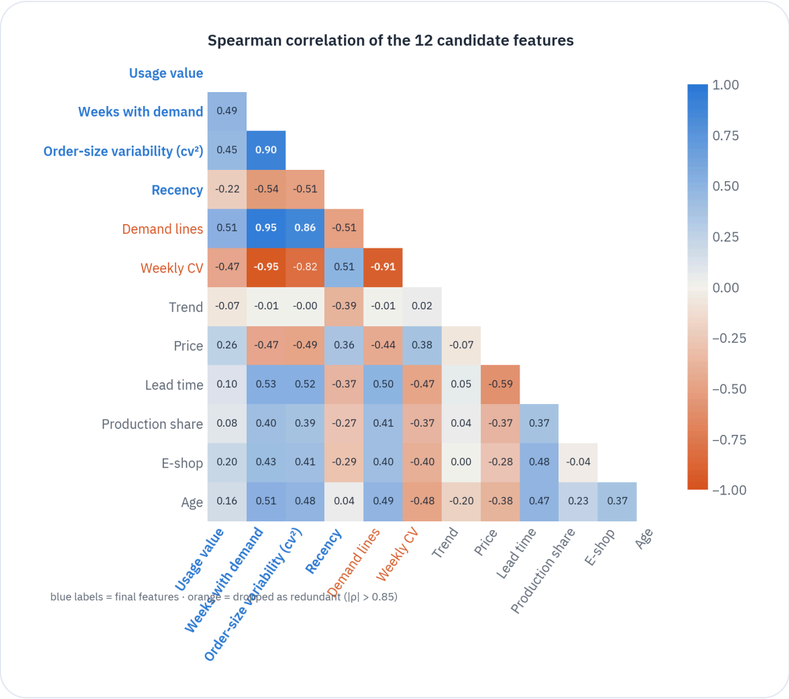
→

Reorder point
+ safety stock

PART 2 · FEATURE SELECTION

From 12 candidate features to 4

Why: keep only features that drive stocking, without overlap



12 candidate features

10 after removing overlaps (|ρ| > 0.85)

4 after business screen

Population: 20,890 articles with demand in the last 3 years, stocked or not

PART 2 · THE FOUR FEATURES

Each class has a clear demand fingerprint

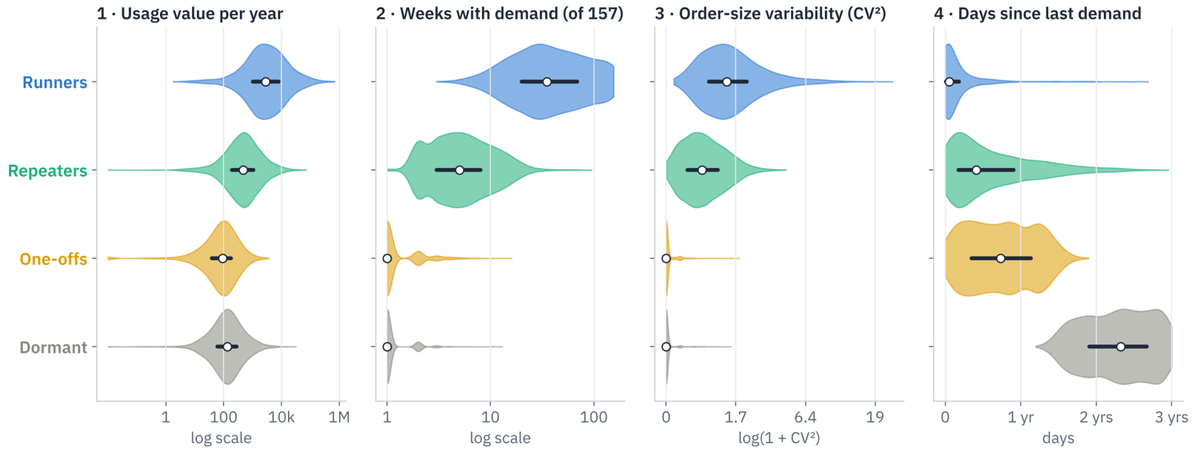
Why: shows the clusters differ for real, on every feature

1 · Usage value
money flowing through it

2 · Weeks with demand
how regularly it is needed

3 · Size variability
how predictable each order is

4 · Recency
is it still alive?

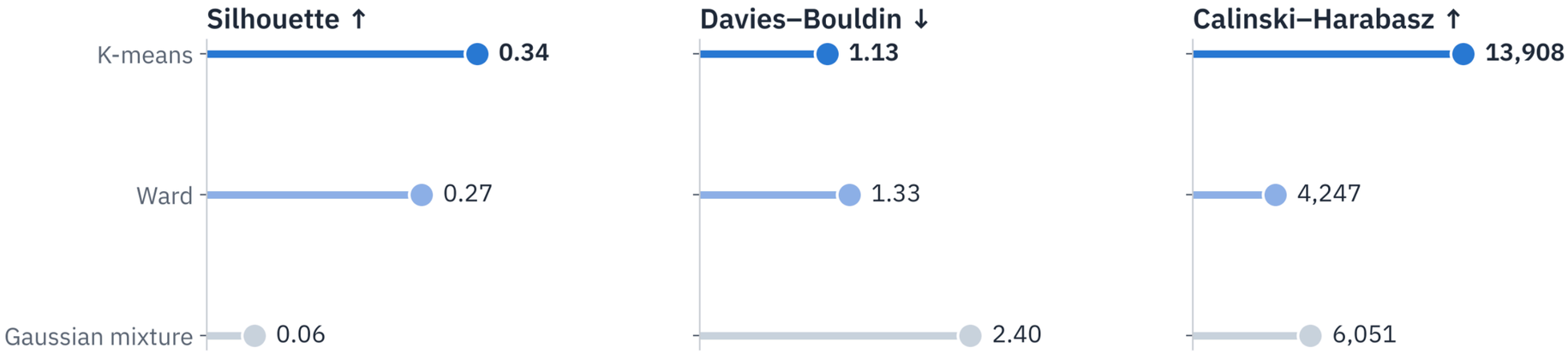


Dot = median · bar = middle 50% of articles

PART 2 · CHOOSING THE ALGORITHM

K-means with 4 classes scores best and is stable

Why: proves the classes are robust, not an artefact



stability across 20 bootstrap refits

0.98

classic ABC-XYZ cannot separate this range

98% Z

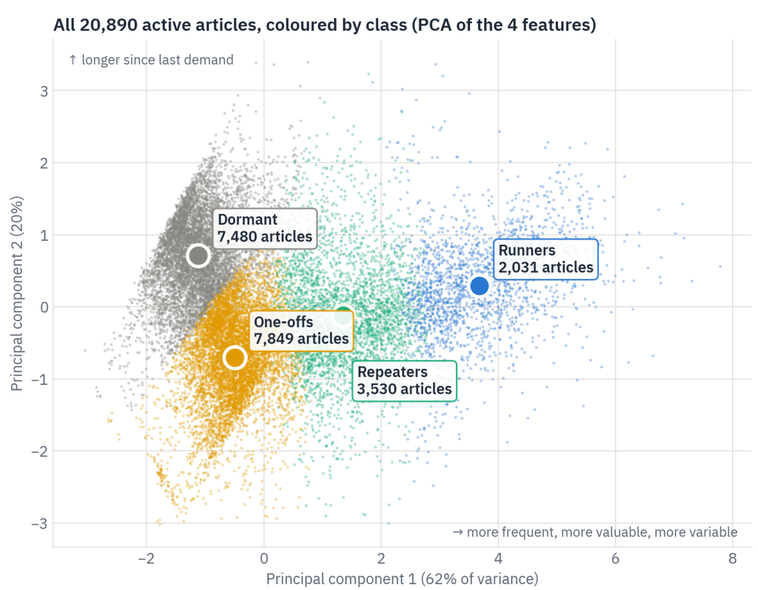
keeps frequent and occasional articles apart

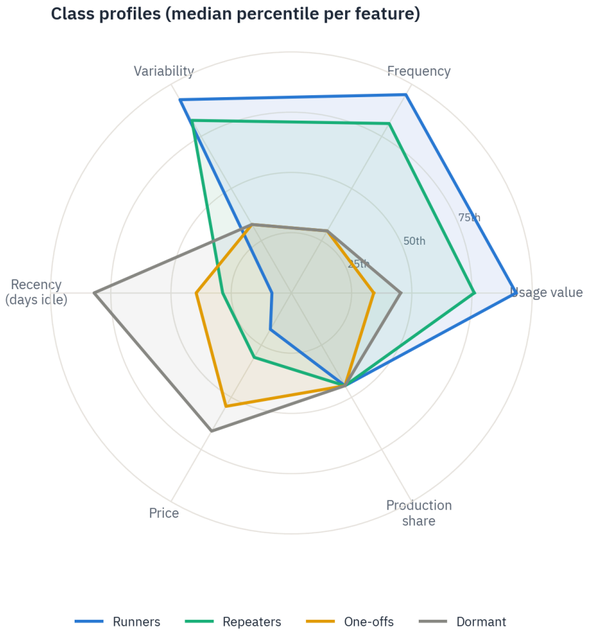
k = 4

Ward on an 8,000-article sample

Four classes, separated by usage and by recency

Why: each dot is one article; where it sits shows how it behaves



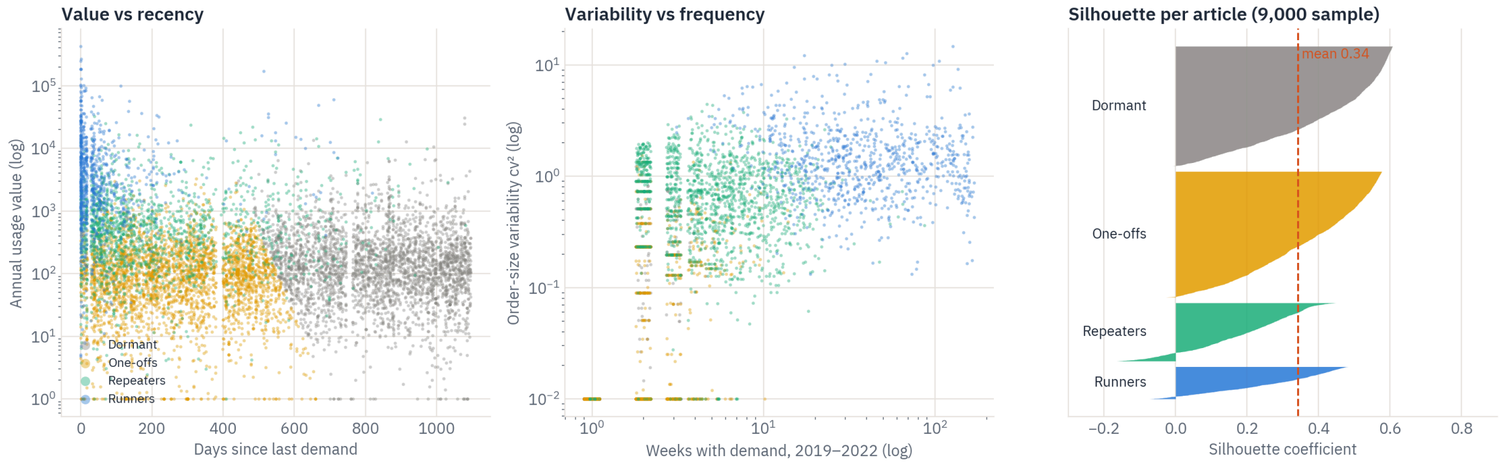


PCA: PC1 62% and PC2 20% of variance

PART 2 · THE CLUSTERS UP CLOSE

Repeaters sit between regular and one-time demand

Why: explains why Repeaters need a selective rule



Runners:
recent and high value

One-offs, Dormant:
a single order

Repeaters:
the transition zone

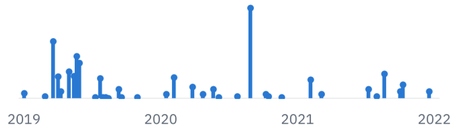
PART 2 · STRATEGY PER CLASS

Different demand, so a different way to supply it

Why: the strategy follows directly from how each class behaves

Runners

10% of articles · 74% of usage



Why: frequent and valuable: a stockout stops sales or production

→ Stock, demand-driven ROP

Repeaters

17% · 14% of usage



Why: a buffer would sit idle most of the year

→ Stock only if supplier is slow

One-offs

38% · 4% of usage



Why: one order in 3 years, expensive per unit

→ Buy to order

Dormant

36% · 8% of usage



Why: no demand for over a year

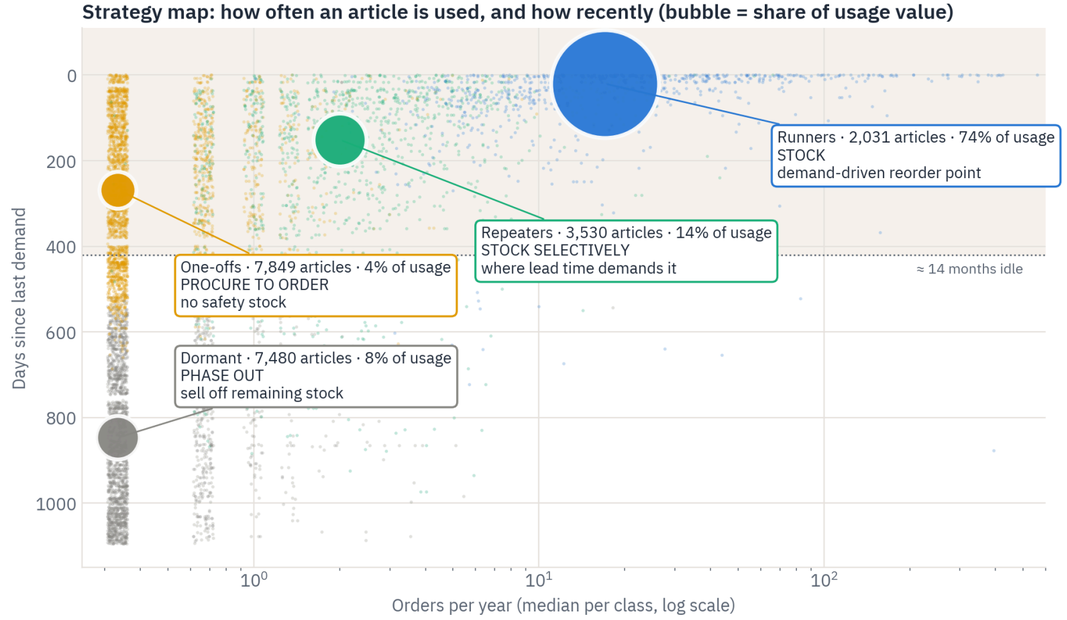
→ Phase out

Strip = weekly demand of one typical article per class, 2019–2022

PART 2 · WHAT CHANGES FOR THE CLIENT

Only Runners and selected Repeaters belong on stock

Why: turns the classes into concrete actions



stock articles to de-list

427

Runners still without a reorder point

1,052

re-classification keeps it current

Quarterly

20,890 active articles · bubble = share of usage value

PART 3 · SERVICE LEVELS

ABC decides how much buffer a stocked article gets

Why: protect the articles where a stockout costs most

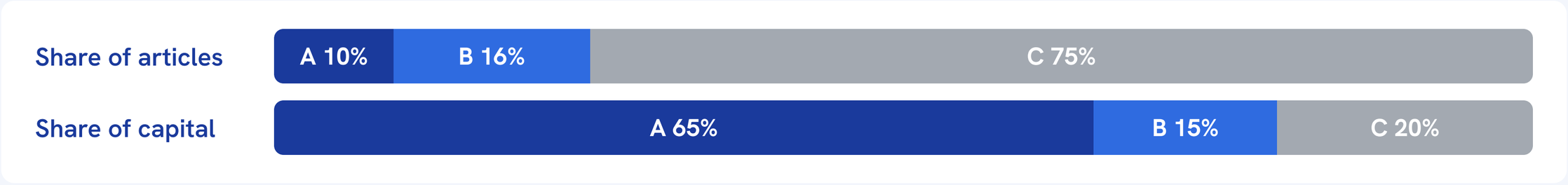
Class	Usage value	Articles	Fill target
A	**top 80%**	**167**	**98%**
B	next 15%	273	95%
C	last 5%	1,311	90%

A-articles: share of articles → share of capital

10% → 65%

each target reached at least capital

Calibrated



12-month usage value at current purchase price · 1,751 inventory articles

PART 3 · WHY SIMULATE

Prove the impact before changing 1,751 settings

Why: a number per article is not proof of what it does

Formulas alone can’t be trusted

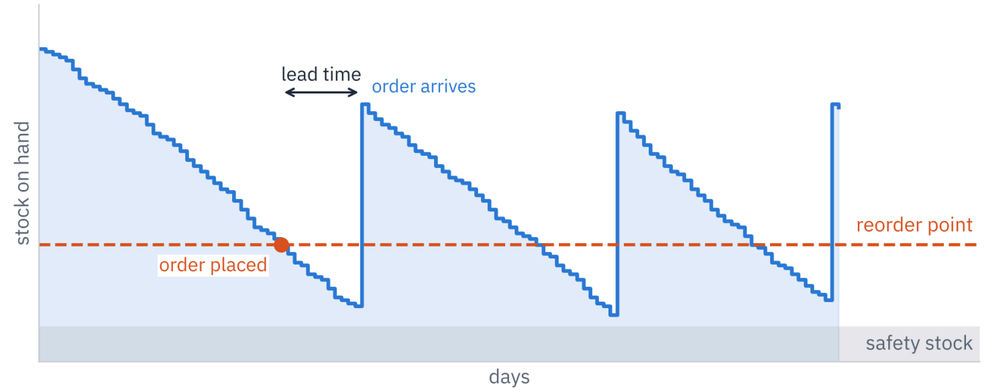
75% irregular demand, large orders, MOQs

History can’t be replayed

planners overrode the settings by hand

Proof before change

test on a year the method never saw



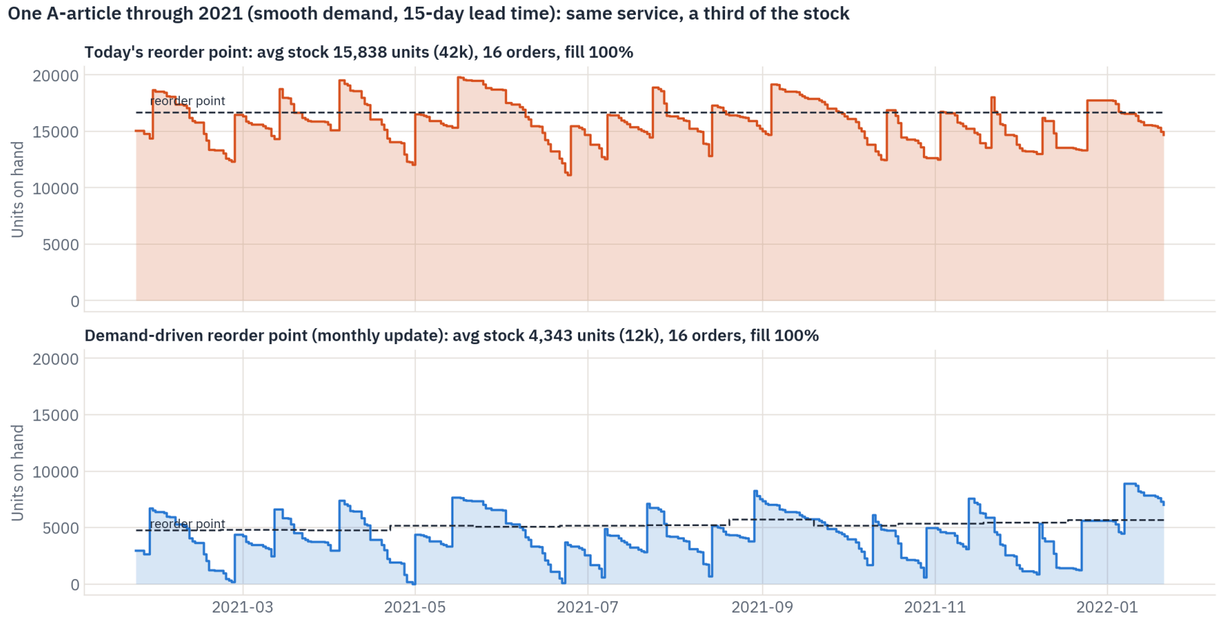
1. real daily demand, 2021
2. order when stock ≤ reorder point
3. delivery after the lead time
4. measure stock and fill

Fit 2019–20 · calibrate 2020 · test 2021 · code: part3_safety_stock_rop/src/sim.py

PART 3 · ONE ARTICLE, SIMULATED

Same orders, same 100% fill, 72% less stock

Why: makes the mechanism visible on a real A-article



today’s reorder point vs need

3.2×

average units on hand

15,800 → 4,300

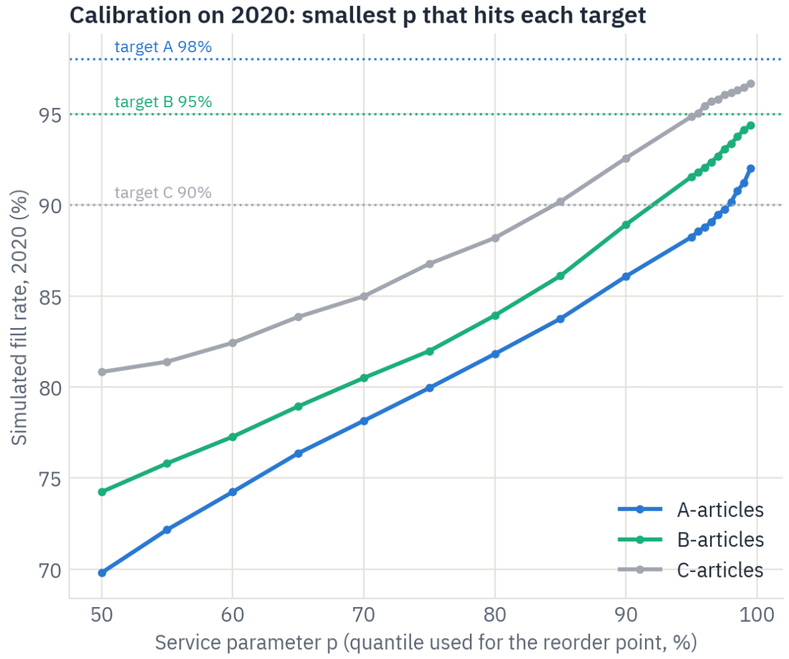
same orders, same 100% fill

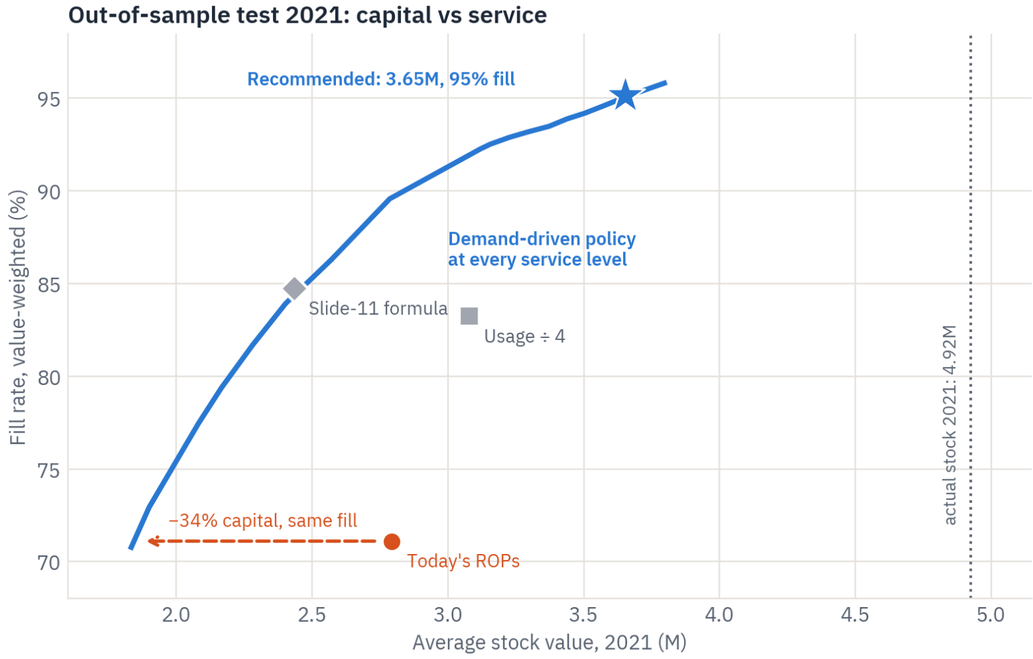
16 orders

PART 3 · CAPITAL VS SERVICE

More service for less capital
than any current rule

Why: lets management pick a service level and see its cost



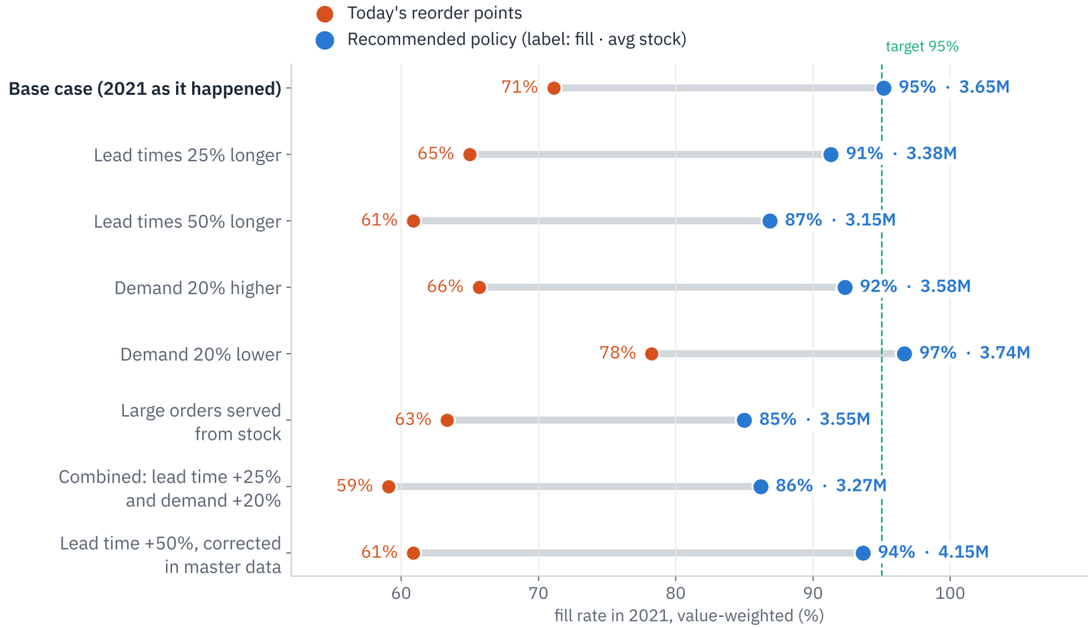


Same simulator, demand, order sizes and lead times for every policy

PART 3 · STRESS TEST

The strategy holds when things go wrong

Why: a recommendation must survive bad lead times and demand shocks



fill in every scenario

≥ 85%

points ahead of today's rules

+18 to +33

the main risk, fixed by correct master data

Lead times

Test year 2021 · demand shocks applied to 2021, reorder points recalculated monthly

PART 3 · WHAT THE SIMULATION GAVE US

From recommendation to evidence

Why: four things a formula alone could not show

71%

Exposed today’s rules

the fill today’s reorder points really deliver

3

Set service levels

targets per ABC class, chosen on 2020 data

−1.27M

Priced the impact

on 2021, a year the method never saw

8

Tested the risks

stress scenarios, lead times first to fix

RESULT — One tested rule replaces 1,751 manual settings, with its cost known in advance

Simulator: daily (s, Q) replay of 1,751 articles in parallel

RESULTS & RECOMMENDED STRATEGY

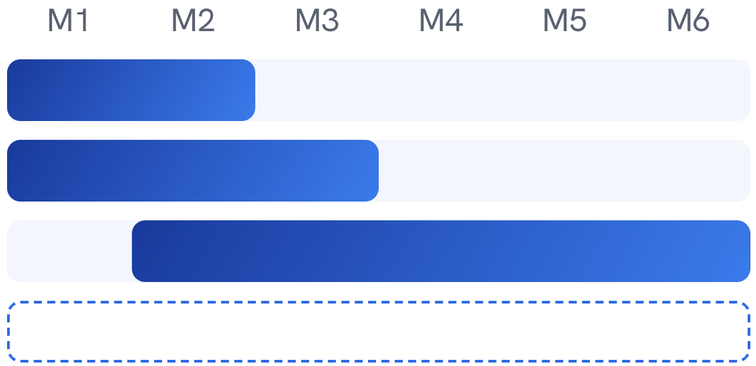
Stock the right articles, at levels set by demand

Policy, 2021	Stock	Fill
Actual 2021	4.92M	–
Today’s rules	2.79M	71%
Usage ÷ 4	3.08M	83%
Recommended	**3.65M**	**95%**

≈ 0.25M a year holding cost saved

–1.27M (–26%)

ROADMAP



Step 3 starts with an A-article pilot · step 4 runs continuously

Per-article levels: Part3_ROP_SafetyStock_Recommendations.xlsx in the repository

LIMITATIONS & NEXT STEPS

Robust to the data gaps, three points to validate

LIMITATIONS

- Master data is a 2022 snapshot
- No order dates to measure lead times
- Holding cost rate of 20% assumed
- Recorded service history is censored

NEXT STEPS

- Agree service targets per class
- Validate A-article lead times
- Run a one-quarter A-article pilot

Thank you · Questions?

Code, notebooks and full report: github.com/divyashori-datascience/scm-inventory-optimisation